QUANTITATIVE METHODS 1

Worksheet 7 – Integrated cases

Detailed solutions

Case A – Detailed solution

Observed population: 200 surveyed students. Unit: one student. Variable: maximum acceptable price, continuous quantitative.

Class	$[5,10[$	$[10,15[$	$[15,20[$	$[20,30[$	$[30,40[$	$[40,60[$
n_i	10	25	45	60	40	20
a_i	5	5	5	10	10	20
d_i	2	5	9	6	4	1
ICC	10	35	80	140	180	200
DCC	200	190	165	120	60	20
c_i	7.5	12.5	17.5	25	35	50

The modal class is $[15,20[$.

$$Q_1 \approx 16.67, \qquad Me \approx 23.33, \qquad Q_3 = 32.50.$$

$$\bar{x} = \frac{5075}{200} = 25.375.$$

$$V = 129.859375, \qquad \sigma \approx 11.396, \qquad CV \approx 44.91\%.$$

For 65% to afford at least price p, use the 35th percentile:

$$0.35(200) = 70,$$

$$p = 15 + 5\frac{70-35}{45} \approx 18.89 \text{ euros}.$$

Satisfied/not satisfied is a binary nominal qualitative variable; a bar chart is appropriate.

Case B – Part 1: regression

$$\bar{x} = 9.125, \qquad \bar{y} = 18.25,$$

$$V(x) = 12.859375, \qquad V(y) = 55.1875, \qquad \mathrm{Cov}(x,y) = 26.59375.$$

$$r \approx 0.99827.$$

$$\hat{y} = 2.06804x - 0.62090, \qquad R^2 \approx 0.99655.$$

Residuals are approximately

$$(0.349, 0.281, 0.145, -0.923, -0.060, -0.128, -0.264, 0.600).$$

$$\hat{y}(18) \approx 36.6039.$$

Since $18 > 15$, the forecast is extrapolation. Strong correlation remains association, not proof of causality.

Case B – Part 2: time series

$$MA_4 : 110,\ 112,\ 114.5,\ 117.5,\ 121.$$

$$CMA : 111,\ 113.25,\ 116,\ 119.25.$$

The series shows an upward trend and recurring quarterly seasonality. Centring is needed because an even-order moving average lies between two periods.

Case B – Part 3: indices

$$I_P^F \approx 122.30, \qquad I_Q^F \approx 107.32, \qquad I_V = 131.25.$$

$$\frac{122.2963 \times 107.3213}{100} = 131.25 = I_V.$$

$$\textit{Real value} = \frac{2.4}{1.20} = 2.0 \text{ million euros.}$$